

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: Book _____		
Property Name	Data Type	Source Tag
_____ BID _____	_____ string _____	_____ System Generated _____
_____ title _____	_____ string _____	_____ User Input _____
_____ author _____	_____ string _____	_____ User Input _____
_____ genre _____	_____ string _____	_____ From List _____
_____ copiesInStock _____	_____ int _____	_____ Default Value _____
_____ isAvailable _____	_____ bool _____	_____ Calculated _____
_____	_____	_____

Entity Name: Member _____		
Property Name	Data Type	Source Tag
_____ MID _____	_____ string _____	_____ System Generated _____
_____ name _____	_____ string _____	_____ User Input _____
_____ dateOfBirth _____	_____ date _____	_____ User Input _____
_____ phoneNumber _____	_____ string _____	_____ User Input _____
_____ email _____	_____ string _____	_____ User Input _____
_____	_____	_____
_____	_____	_____

Entity Name: Librarian		
Property Name	Data Type	Source Tag
LID	string	Computer Generated
name	string	User Input
phoneNumber	string	User Input
email	string	User Input
role	string	Default Value

Entity Name: BorrowedBook		
Property Name	Data Type	Source Tag
BBID	string	SystemGenerated
member	string	Calculated
book	string	Calculated
MaxDays	int	Default Value
borrowDate	date	Default Value
overdueDays	int	Calculated
fine	decimal	Calculated
status	string	Default Value

Entity Name: Reservation _____		
Property Name	Data Type	Source Tag
RID	string	System Generated
member	string	Calculated
book	string	Calculated
reservationDate	date	Default Value
status	string	Default Value
isNotified	bool	Default Value
_____	_____	_____

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

The system allows a member to register (Member entity created) and/or librarian (Librarian entity) to login. A book can be added to the library (Book entity created). Member can borrow a book only if the book is available in stock (new BorrowedBook entity created) and the Book entity's copiesInStock will get decreased in order, set borrowDate and status. Then, if there are no more copies (copiesInStock == 0); isAvailable property get switched to false. In addition, a book can be reserved by a member (create new Reservation) and will get notified once the book is available.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterNewMember	Create new Member entity
2	AddBook	Add new book to the library
3	AddCopy	Add copy of an existing book to the library
4	DeleteBook	Delete a book from the library system (or set as unavailable)
5	UpdateMember	Update member information
6	DeleteMember	Delete member from the system
7	BorrowBook	Add BorrowedBook entity; Assign all details
8	ReserveBook	Add Reservation entity; assign all details
9	ViewAllborrows	Display all BorrowedBook entities with details
10	ViewAllReservations	Display all Reservation entities with details

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	
guestName	string	
guestPhone	string	
guestEmail	string	
guestNationality	string	
registrationDate	string	

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	_____
roomNumber	string	_____
roomType	string	_____
floorNumber	int	_____
nightlyPrice	decimal	_____
isAvailable	bool	_____

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	_____
guestId	int	_____
roomId	int	_____
checkInDate	string	_____
checkOutDate	string	_____
numberOfNights	int	_____
totalPrice	decimal	_____
status	string	_____

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	_____
bookingId	int	_____
guestId	int	_____
roomId	int	_____
actualCheckIn	string	_____
actualCheckOut	string	_____
finalAmount	decimal	_____

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____
5	_____	_____
6	_____	_____
7	_____	_____

#	Service Name	What it does
8		
9		